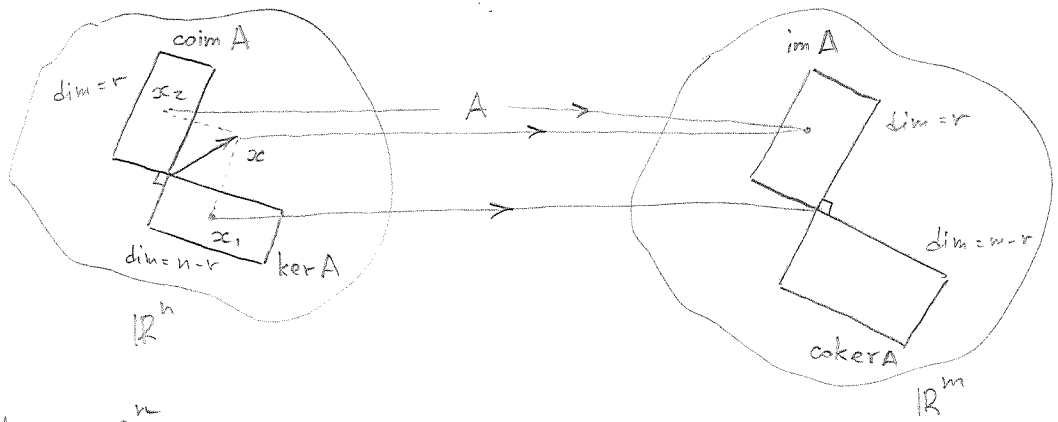
Lecture 11

- Geometry of $\mathbb{R}^n \xrightarrow{A} \mathbb{R}^m$
- Eigenvalues and Eigenvectors
- Gerschgorin Theorem
- Eigen Bases

The Fundamental Theorem of Linear Algebra allows us to understand the geometry behind a linear map $\mathbb{R}^n \longrightarrow \mathbb{R}^m$ induced by a matrix $A \in \mathbb{M}_{m\times n}$



Let $x \in \mathbb{R}^n$

- $\exists!\ x_1 \in \ker A$ and $x_2 \in \operatorname{coim} A$: $x = x_1 + x_2$

$$x_1 = \operatorname{proj}_{\ker A} x \quad , \quad x_2 = \operatorname{proj}_{\operatorname{coim} A} x$$

- $Ax = A(x_1 + x_2) = Ax_2 = b$

Map $\quad x \xmapsto{pr} \underset{\operatorname{coim} A}{\underset{\cap}{x_2}} \longmapsto \underset{\operatorname{im} A}{\underset{\cap}{b}}$

- Mapping $\operatorname{coim} A \xrightarrow{A} \operatorname{im} A$ is <u>1-to-1 correspondence</u>

$\forall b \in \operatorname{im} A\ \exists!\ x_2 \in \operatorname{coim} A$ such that $Ax_2 = b$.

Indeed:

(a) $\forall b \in \operatorname{im} A\ \exists x_2 \in \operatorname{coim} A : Ax_2 = b$.

This is true since $\forall b \in \operatorname{im} A\ \exists x \in \mathbb{R}^n : Ax = b$, take $x_2 = \operatorname{proj}_{\operatorname{coim} A} x$.

(b) Uniqueness

Suppose $x_1, x_2 \in \operatorname{coim} A$: $Ax_1 = b$, $Ax_2 = b$

$\Rightarrow A(x_1 - x_2) = 0 \Rightarrow x_1 - x_2 \in \ker A$

$\Rightarrow x_1 = x_2$

<u>Thm</u> Let $Ax = b$ be a compatible system ($b \in \operatorname{im} A$)

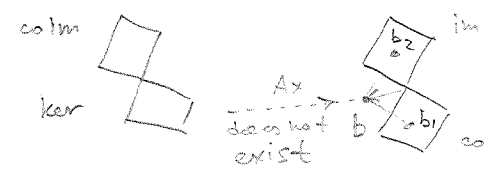
- $\exists!\ x^* \in \operatorname{coim} A : Ax^* = b$
- General solution $x = x^* + \ker A$

<u>Remark</u> Particular sol. $x^* \in \operatorname{coim} A$ has an important property. it is a sol. with the smallest Euclidean norm:

$$\|x\|^2 = \|x^* + x_{\ker}\|^2 = \|x^*\|^2 + \|x_{\ker}\|^2 \geq \|x^*\|^2.$$

any solution

<u>Q</u>: What if $b \notin \operatorname{im} A \Rightarrow x^* = \arg\min_x \|Ax - b\|^2$



$b = b_1 + b_2$, $b_1 \in \operatorname{coker}$, $b_2 \in \operatorname{im}$

$Ax^* = b_2 = pr_{\operatorname{im} A} b$

$b_1 = b - b_2 \in \operatorname{coker} A$

$A^T b_1 = 0 \Rightarrow \boxed{A^T b = A^T A x^*}$

normal equations.

Eigenvalues and Eigenvectors

Let $A \in \mathbb{M}_{n\times n}$.

Def A scalar λ is called an eigenvalue of A if there exist $v \in \mathbb{R}^n$, $v \neq 0$, such that

(1) $\boxed{Av = \lambda v}$ ← epic equation of LA.

Def Any vector $v \in \mathbb{R}^n$, $v \neq 0$, satisfying (1) is called an eigenvector associated with eigenvalue λ.

(1) is a nonlinear (in λ and v) system of equations.

We can rewrite it equivalently:

(2) $\boxed{(A - \lambda I_n)v = 0}$, I_n is $n\times n$ identity matrix.

Key observation: for given λ, (2) has a nonzero solution $\iff$ $(A - \lambda I_n)$ is singular.

Strategy:

① Find all λ for which $(A - \lambda I)$ is sing:

$$\underbrace{p_A(\lambda)}_{\text{characteristic polynomial } (\deg = n)} = \det(A - \lambda I) = 0$$

② For each λ, solve $(A - \lambda I)v = 0$ for v.

Fund Thm of Algebra: $p_A(\lambda) = (-1)^n(\lambda - \lambda_1)\dots(\lambda - \lambda_n)$ (*),

where $\lambda_1 \dots \lambda_n$ are the roots of $p_A(\lambda)$ = eigenvalues of A

In general, $\lambda_i \in \mathbb{C}$.

For most matrices, $\lambda_1 \dots \lambda_n$ are distinct.

But in general, A may have repeated eigenvalues.

Def An eigenvalue λ_i is said to have algebraic multiplicity m_i if the factor $(\lambda - \lambda_i)$ appears m_i times in (*)

$$p_A(\lambda) = (-1)^n \dots (\lambda - \lambda_i)^{m_i} \dots$$

Def If $m_i = 1$, then λ_i is called simple eigenvalue

Let $\lambda \in \operatorname{spec}(A)$ (be an eigenvalue of A).

Then $\boxed{V_\lambda = \ker(A - \lambda I)}$ is a subspace of $\mathbb{R}^n$ if $\lambda \in \mathbb{R}$, of $\mathbb{C}^n$ if $\lambda \in \mathbb{C}$, spanned by all eigenvectors assoc. with λ.

Remark: $\lambda \in \operatorname{spec}(A) \iff V_\lambda \neq \{0\}$.

Def $\dim V_\lambda$ is called the geometric multiplicity of λ.

Remark If $\lambda = \mu + i\nu \in \operatorname{spec}(A)$ and $v = a + ib \in V_\lambda$

then $\bar{\lambda} = \mu - i\nu \in \operatorname{spec}(A)$ and $\bar{v} = a - ib \in V_{\bar{\lambda}}$.

$Av = \lambda v \Rightarrow \overline{Av} = \overline{\lambda v} \Rightarrow A\bar{v} = \bar{\lambda}\bar{v}$.

Remark In practice, Strategy ①, ② is not used.

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1. Finding det is expensive
2. Finding roots of polynomials is a nontrivial task.

Gerschgorin Theorem

Let $A \in \mathbb{M}_{n\times n}$ (real or complex).

Let $D_i = \{z \in \mathbb{C} : |z - a_{ii}| \le r_i\}$, $r_i = \sum_{j \ne i} |a_{ij}|$

Gerschgorin disk

Let $D_A = \bigcup_{i=1}^{n} D_i \subset \mathbb{C}$

Gerschgorin domain.

Thm $\underbrace{\mathrm{spec}(A)}_{\text{set of all eigenvals of } A} \subset D_A$.

It has one useful corollary.

Recall that A is singular $\iff$ $\lambda = 0$ is an eigenval.

Therefore: if $0 \notin D_A \Rightarrow A$ is nonsing.

Eigen Bases

It turns out that for most of the matrices we can construct a basis of $\mathbb{R}^n$ (or $\mathbb{C}^n$), consisting of eigenvectors of A.

Lemma: If $\lambda_1 \dots \lambda_k$ are distinct eigenvalues of A and $v_1 \dots v_k$ are the eigenvectors assoc with λ_i

$\Longrightarrow$ $v_1 \dots v_k$ are lin independent.

Proof: Induction on k.

1. If $k = 1 \Rightarrow (\lambda_1, v_1)$, $v_1 \ne 0$ (by def).
2. Suppose the statement is true for $(k-1)$.
3. Let's show it is true for k

Suppose $v_1 \dots v_k$ are lin. dependent:

$\alpha_1 v_1 + \dots + \alpha_k v_k = 0$ (we want to show that all $\alpha_i = 0$)

$\Rightarrow A(\ ) = \alpha_1 \lambda_1 v_1 + \dots + \alpha_k \lambda_k v_k = 0$

$\Rightarrow \lambda_k (\ ) = \alpha_1 \lambda_k v_1 + \dots + \alpha_k \lambda_k v_k = 0$ } "—"

$\alpha_1 (\lambda_1 - \lambda_k) v_1 + \dots + \alpha_{k-1} (\lambda_{k-1} - \lambda_k) v_{k-1} = 0$

2) $\Rightarrow \begin{cases} \alpha_1 (\lambda_1 - \lambda_k) = 0 \\ \vdots \\ \alpha_{k-1} (\lambda_{k-1} - \lambda_k) = 0 \end{cases} \Rightarrow$ since λ_i are distict $\alpha_i = 0$, $i = 1, \dots, k-1$ $\Rightarrow \alpha_k = 0$ since $v_k \ne 0$. $\square$

This lemma has an immediate corollary:

Thm If all eigenvalues of A are distinct and real, then the assoc. eigenvectors $v_1 \dots v_n$ form a basis (eigenbasis) of $\mathbb{R}^n$

—"— distinct and complex $\Rightarrow$ —"— $\mathbb{C}^n$.

Remark Let $\{v_i\} = \underbrace{\{u_i\}}_{\text{real}} \cup \underbrace{\{w_i\}}_{\text{complex}}$ form a basis of $\mathbb{C}^n$ $\quad w_i^{\pm} = a \pm bi$

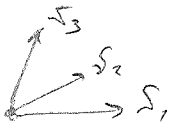
$\Rightarrow \{u_i\} \cup \{\text{Real part of } w_i\} \cup \{\mathrm{Im}\, w_i\}$ form a basis in $\mathbb{R}^n$.

Q What if not all $\lambda_i \in \operatorname{spec}(A)$ are distinct?

Then A may or may not have an eigenbasis.

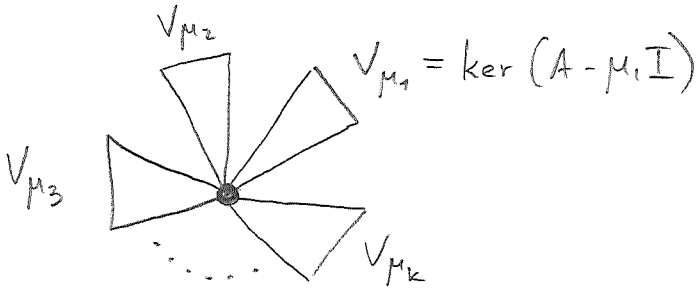
Let $\mu_1, \dots, \mu_k$ denote distinct values of eigenvalues $\lambda_1 \dots \lambda_n$ (e.g. $\lambda_1 = \lambda_2 = \mu_1$, $\lambda_3 = \lambda_4 = \lambda_5 = \mu_2$, ect.), $k \le n$

- If $k = n \Rightarrow$ all $\lambda_1 \dots \lambda_n$ are distinct.



$v_1 \quad v_n$ eigenbasis

- If $k < n \Rightarrow$ repeated eigenvalues



To construct an eigenbasis (basis, consisting of vectors in $V_{\mu_1}, \dots, V_{\mu_k}$), we need to take

$$\underbrace{\{\text{Basis of } V_{\mu_1}\} \cup \dots \cup \{\text{Basis of } V_{\mu_k}\}}_{\text{are lin. independent.}} \quad (\#)$$

vectors : $\dim V_{\mu_1} + \dots + \dim V_{\mu_k}$.

Key fact : $\boxed{\dim V_{\mu_i} \le m_i}$

Therefore,

$\#\text{vectors} \le m_1 + \dots + m_k = n$.

Collection of vectors $(\#)$ is a basis $\Leftrightarrow \dim V_{\mu_i} = m_i$

Def An eigenvalue is called complete if $\dim V_\lambda = m_\lambda$

Def A matrix is complete if all its eigenvalues are complete.

Thm A has an eigenbasis $\Leftrightarrow$ A is complete

Example

- $A = \begin{bmatrix} a & & 0 \\ & \ddots & \\ 0 & & a \end{bmatrix} \Rightarrow \lambda = a \in \operatorname{spec}(A)$ $\quad m_{\lambda=a} = n$, $V_{\lambda=a} = \mathbb{R}^n$, $\dim V_{\lambda=a} = n$

- $A = \begin{bmatrix} a & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & a \end{bmatrix} \Rightarrow \lambda = a \in \operatorname{spec}(A)$ $\quad m_{\lambda=a} = n$, $V_{\lambda=a} = \operatorname{span}(e_1)$, $\dim V_{\lambda=a} = 1$.

Jordan block